Process Engineering Equipment Handbook

Claire Soares

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Abrasive Technology, Inc.: 8400 Green Meadows Drive, Westerville, OH 43081 USA
Sandusky-Chicago Abrasive Wheel Co., Inc.: 1100 W. Barker Avenue, Michigan City, IN 46360 USA
National Metal Abrasive, Inc.: P.O. Box 341, Wadsworth, OH 44282 USA

Acid Rain

Environment Canada: Environment Canada Enquiry Centre, Ottawa, Ontario K1A 0H3 Canada

Acoustic Enclosures, Turbine

Altair Filters International Limited: Omega Park, Alton, Hampshire GU34 2QE England

Actuators

J.M. Voith GmbH: P.O. Box 1940, D-89509 Heidenheim, Germany
Voith Turbo GmbH & Co., KG: P.O. Box 1555, D-74555 Crailsheim, Germany

Air Filtration

Altair Filters International Limited: see above

Air Pollution Control

Alstom (formerly ABB Power Generation): Finspong 61282 Sweden; Hasselstrasse 16, CH-5401 Baden, Switzerland; 5309 Commonwealth Center Parkway, Midlothian, VA 23112 USA

B

Bearings

Demag Delaval: 840 Nottingham Way, Trenton, NJ 08638 USA
Revolve Magnetic Bearings, Inc.: Calgary, Alberta, Canada

Boilers

Environment Canada: see above

Brakes

J.M. Voith GmbH: see above

C

CFD (Computational Fluid Dynamics)

Fluent Inc.: Lebanon, NH USA

Carbon; Carbon-Graphite Mix Products

Advance Carbon Products: 2036 National Avenue, Hayward, CA 94545 USA

Carbon Dioxide (CO_2); CO_2 Disposal

Vatenfall: S162 87 Stockholm, Sweden

Cement; Portland Cement

Environment Canada: see above

Centrifuges

Dorr-Oliver Inc.: 612 Wheelers Farm Road, Milford, CT 06460 USA

Chemical Complex; (Petro)Chemical Complex; Chemical Plant

Petrochemcial Company of Singapore: Singapore

Chemicals

ARCO Chemical Company: 3801 West Chester Avenue, Newton Square, PA 19073-2387 USA

Chemicals (Toxic), Handling

ARCO Chemical Company: see above

Chillers; Crystallizers; Chemical Separation Method; Alternative to Distillation/Fractional Distillation

Armstrong Engineering Associates: P.O. Box 566M, West Chester, PA 19381-0566 USA

Cogeneration

Alstom: Finspong 61282 Sweden

Compressors

Sulzer-Burckhardt: Winterthur, Switzerland
Demag Delaval: see above
Aerzener Maschinenfabrik, GmbH: D2358 Aerzen, Germany

Condensers

Alstom: Hasselstrasse 16, CH-5401 Baden, Switzerland; D6800 Mannheim 1, Germany

Condition Monitoring (CM); Condition-Monitoring System(s) (CMS); Engine Condition Monitoring (ECM); Engine Condition–Monitoring System(s) (ECMS)

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Control Systems; Controls

Sulzer-Burckhardt: see above
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Voith Safeset A.B.: Ronningev. 6, S-82434 Hudliksvall, Sweden

Controls, Retrofit

Petrotech Inc.: 108 Jarrel Drive, P.O. Box 503, Belle Chase, LA 70037 USA
Demag Delaval: see above
J.M. Voith GmbH: see above

Conveyors

Sandvik Process System, Inc.: USA
Bloch, H., and Soares, C. M., *Process Plant Machinery*, 2d ed., Butterworth-Heinemann, 1998.

Coolant; Engine Coolant

ARCO Chemical Company: see above

Cooling; Cool, Products That (Air Conditioners); Liquid-Cooled Air Conditioners

Thermoelectric Cooling America Corporation (TECA): USA

Cooling Towers

The Marley Cooling Tower Company: Marley and Lone Elm Roads, Olathe, KS 66061 USA

D

Drying

ARCO Chemical Company: see above

E

Ecological Parks; Industrial Ecological Parks

Environment Canada: see above

Ecosystem

Environment Canada: see above

Electric Motors; Electric Motor Controls

Reliance Electric Company: Cleveland, OH USA

Emissions; Air Emissions

Environment Canada: see above

Engines, Gas

Cooper-Bessemer Reciprocating: Grove City, PA USA

Environmental Accountability

Kodak: USA
Cultor: Finland

Environmental Economics

AssiDomän: Sweden

Exhausters, Centrifugal Gas

Ansaldo: Milan, Italy

Expansion Joints

Townson Expansion Joints: United Kingdom

Explosion; Explosion Hazard Analysis; Explosion Hazards

Eutech Engineering Solutions Ltd.: Billingham, Cleveland TS23 4YS England
H.M. Principal Specialist Inspector, Health & Safety Executive, Quay House, Quay Street, Manchester M3 3JB England

F

Fans, Centrifugal

Ansaldo: see above
Bloch, H., and Soares, C. M., *Process Plant Machinery*, 2d ed., Butterworth-Heinemann, 1998.

Filters

Peerless Manufacturing Company: 2819 Walnut Hill Lane, Dallas, TX 75229 USA

Forest Products

AssiDomän: see above

Fuel Gas Conditioning System(s)

Peerless Manufacturing Company: see above

Fuel Systems; Fuel Flow Control

J.M. Voith GmbH: see above
Whittaker Controls, Inc.: 12838 Saticoy Street, North Hollywood, CA 91605 USA

Fuels, Alternative; Fuels, Gas Turbine

Bechtel Power Corporation: Gaithersburg, MD 20878 USA

G

Generators; Turbogenerators

Alstom: see above

H

Heat Exchangers

Armstrong Engineering Associates: see above

Heat Pumps; Heat Pumps, Geothermal; Heating Systems with a Renewable Energy Source

Enertran: Canada

L

Life-Cycle Assessment (LCA) (of Turbomachinery)

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Liquid Natural Gas (LNG)

Peerless Manufacturing Company: see above

Lubrication

Demag Delaval: see above

M

Measurement

Demag Delaval: see above

Metallurgy; Metallurgical Repair; Metallurgical Refurbishment

Liburdi Engineering: Hwy. 400, Dundas, Ontario, Canada

O

Oil Sands; Synthetic Crude; Tar Sands; Shale

Syncrude Canada Limited: Ft. McMurray, Alberta, Canada

Oxygen Analysis

Rosemount Analytical: Orville, OH USA

Ozone

Environment Canada: see above

P

Pollutants, Chemical; Pollutants, (from) Chemical Processes; Pollutant Indicators; Pollutants, Toxic; Pollutants, Toxic Chemicals

Environment Canada: see above

Power Transmission

Demag Delaval: see above
MAAG Gear Company: Switzerland
J.M. Voith GmbH: see above

Pulp and Paper

AssiDomän: see above

Pulsation Dampeners

Peerless Manufacturing Company: see above

Pumps

Bloch, H., and Soares, C. M., *Process Plant Machinery*, 2d ed., Butterworth-Heinemann, 1998.
Demag Delaval: see above
Sulzer Pumps: USA
Goulds Pumps: USA

R

Refineries, Petroleum

Environment Canada: see above

S

Seals; Gas Seals

Revolve Magnetic Bearing, Inc.: see above

Separators

Peerless Manufacturing Company: see above

Stacks

Altair Filters International Limited: see above

T

Tanks

A.O. Smith Engineered Storage Products Company: 2101 S. 21st Street, Parsons, KS 67357 USA
Enraf: England

Turbines, Gas

Alstom: see above
ASME: see above

Turbines, Steam

Demag Delaval: see above
Peerless Manufacturing Company: see above

Turbochargers

Demag Delaval: see above

Turboexpanders

Demag Delaval: see above

U

Ultrasonic Cleaning

Sonics: USA

V

Vaporizers; Vaporizor Applications

Armstrong Engineering Associates: see above

W

Waste Management

Environment Canada: see above

About the Author

A registered professional engineer in Texas and Alberta, Canada, Claire Soares graduated with a B.Sc.Eng. in 1972 and an M.B.A. in 1993. Her career began in computational fluid dynamics working for Brian Spalding in Imperial College, London, on the COBALT project. She then spent about two years working on developing structural patents for the marine and the power distribution industries in England and Canada. Her rotating machinery career began in earnest at the oil sands Syncrude site in Fort McMurray, Alberta, in 1975. Four years later, she moved to Esso Resources and conventional oil and gas production until the "oil patch sat on its tail" at the end of 1981. She then accepted a three-year commission with the Canadian Air Force as Propulsion Systems Manager for all transportation engines in the Department of Defence Transport Command in Canada. She took charge of six helicopter engine fleets, as well as projects related to selecting replacements for two of those fleets. After that, she moved to the United States to start work as a senior engineer for Ryder Airline Services Division (ASD was also called Aviall and, before that, CooperAirmotive). At that point ASD was the largest independent overhaul facility for airline engines in the world, with a shop capacity of about 1000 engines a year. She ran technical support on second shift for 250 mechanics and their supervisors on the JT8D, JT3D, and CFM 56 engine lines. Three years later, she was made manager of the V2500 engine repair program, at that time the first and only designated facility for this engine in the United States. Two years later, after the engine line was up and running, she left to become an independent consultant, trainer, and writer. She has now lived and worked on four continents. Ms. Soares organizes one to two conference sessions annually for the International Gas Turbine Division of the American Society of Mechanical Engineers (ASME) and has done so since 1985. In May 2001 she was appointed to Fellow grade by ASME.

Process Engineering Equipment Handbook is her fourth book. The first, *Process Plant Machinery*, Second Edition, was coauthored with Heinz Bloch, P.E. This work and *Environmental Technology and Economics: Sustainable Development in Industry* helped provide the present handbook with its broad perspective. *Turboexpanders and Process Applications*, her third book, also coauthored with Heinz Bloch, was released in 2001. All her books are used when appropriate as course instruction material for her own and others' courses. Ms. Soares writes extensively for technical journals, such as *Petroleum Economist*, *Asian Electricity*, and *International Power Generation.*

She also writes for more general audiences, with some television screenplays and articles for various international newspapers and magazines to her credit. A published photographer, she writes poetry and has staged multimedia performances of her work for organizations such as the city of Dallas. She is a scuba diver and licensed commercial pilot, and also enjoys swimming and hiking.